Subsea Integrity Copilot - Customer Brief

The value proposition. Keep the pressure alerts; improve the investigation that follows. Help an integrity engineer answer: "Leak or faulty sensors?", "Where should we inspect, and why?" and "What evidence supports this, and what remains unknown?"

Customer and problem. NorthSea Integrity Operations is a fictional offshore gas operator. Its sensors stream pressure readings along a long pipeline network; existing anomaly detectors flag unusual values. Engineers must then compare nearby and paired instruments, source times, pipeline topology and operating records. A leak can produce related pressure changes around a section, while faulty instruments can produce scattered spikes, gaps or drift. But operating changes and shared equipment failures can also affect several sensors, and leaks and faults can coexist. An anomaly is therefore a starting point, not a diagnosis. This manual investigation concentrates work in a few experienced engineers and makes knowledge difficult to transfer. Missed leaks risk environmental releases, disruption and regulatory exposure; false diagnoses can mobilize costly vessel or ROV inspections.

Why I chose this problem. My PhD studied marine data quality, anomaly detection and translating expert requirements into reviewable software. This applies that experience to a high-consequence decision where trusting the measurements matters as much as detecting change.

Same measurements and alerts; different investigation workflows. Both modes use the same scenario, seed, measurement window, detector warnings and inspection interface. The alert-only mode illustrates the prospect's assumed current process, not a measured customer installation.

Sensor alerts only	Agent-assisted investigation
Engineers see individual warnings and manually connect sensor timing, locations, paired readings and operating records. They assemble the diagnosis, inspection priority and handoff themselves. No agents or model calls run.	Agents investigate sensor quality, compare pressure histories and read incident-time records. They provide cited findings, competing explanations and a supported inspection interval, or explain why one is withheld. A grounding reviewer checks source support; rejected reports are withheld.

What we will demonstrate. Pin the same S1 source window in both modes. First show what the engineer must reconstruct from individual alerts; then open the agent-assisted findings, measured comparisons, cited documents and next verification steps. Contrast scattered instrument faults and a mixed leak/fault case. The added value is a reviewable investigation and handoff, not extra alarms. Numerical warnings never wait for AI, and humans retain the decision. Evaluation checks findings against synthetic labels authored before inference, retaining failures, misses and false alarms. Answer-quality ratings are advisory model opinions. Saved Detection playback makes no new model calls; fresh investigations show actual latency.

The two customer roles and objections. The **Head of Subsea Operations** owns the budget and asks whether lower investigation effort justifies integration, inference and review costs without increasing operational risk. The **Lead Subsea Integrity & Instrumentation Engineer** asks whether conclusions survive missing data, inconsistent units, conflicting documents, shared failures and reproduction from source evidence. They have not signed because a cheaper fixed workflow may suffice, AI can be wrong, and data movement and accountability need approval.

The offer and buying gate. Agree a proposed four-week, read-only shadow pilot on one pipeline, subject to data/security access. Establish a manual/fixed-workflow baseline, an inference spend cap and acceptance thresholds before starting. Measure engineer review time, missed/false findings, useful inspection intervals, unsupported claims, abstention and end-to-end availability/latency, including failed attempts. Expand only if evidence quality, effort and total cost justify adoption; no savings or response-time guarantee is assumed.

Assumptions and limits. Data and records are synthetic; pressure behavior is not a validated hydraulic model. Inspection intervals are priorities, not confirmed leak coordinates or dispatch authorization. Hosted inference needs approval; production connectors and expert-approved procedures require customer work. There is no independent engineering validation or field-reliability claim. Humans retain safety authority.

Demo and run instructions: github.com/ntnguyen-so/subsea-integrity-copilot.